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# ELECTROENCEPHALOGRAPHIC BIOMARKERS FOR POST-TRAUMATIC STRESS DISORDER DIAGNOSIS USING EXPLAINABLE ARTIFICIAL INTELLIGENCE

## INTRODUCTION:

- PTSD affects ~3.9% of the global population; diagnosis is often subjective, costly, or invasive. Traditional tools like clinical interviews, fMRI, and questionnaires have limitations in scalability and neurobiological accuracy.
- EEG is a promising non-invasive, portable, and affordable technique that captures brain activity in real time.
- **Challenge:** EEG data is high-dimensional and difficult to interpret manually.
- **Solution:** Combine EEG with Machine Learning (ML) and use dimensionality reduction for data-driven diagnosis.
- **Problem:** ML models are often black-box, limiting clinical trust.
- **Our Approach:** Integrate Explainable AI (XAI) tools – SHAP, LIME, ELI5 – for transparency and interoperability.
- **Key Findings:**
  - Beta Power in the Right Primary Motor Cortex and IQ are crucial biomarkers.
  - ElasticNet feature selection reduces EEG data by ~97% while preserving key signals.
- **Impact:** A clinically feasible, interpretable, and equitable framework for PTSD diagnosis using EEG + ML + XAI.

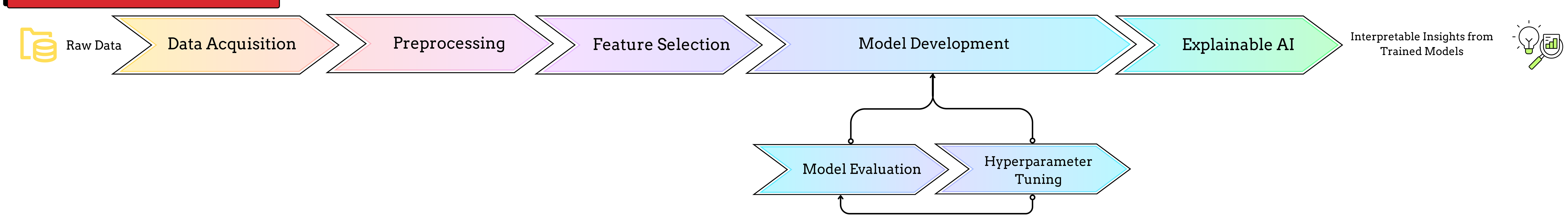
## LITERATURE REVIEW:

- Numerous studies have explored PTSD diagnosis using diverse data types such as EEG, fMRI, structural MRI, linguistic data, and multimodal biometric inputs.
- ML algorithms, including SVM, Random Forest, and Deep Learning, achieved generally high accuracy (above 70%) across data types [1].
- However, the majority of these models operate as black boxes, providing limited clinical transparency and feature-level insight. Only a few studies have begun integrating explainable AI (XAI) tools like SHAP or LIME, and almost none in EEG-based approaches.

**Research Gaps Identified**

- Limited use of XAI in EEG-based PTSD diagnostics
- Underexplored EEG biomarkers in interpretable models
- Need for clinically valid, fair, and explainable AI tools for psychiatric diagnosis

## METHODOLOGY:




## IMPLEMENTATION DETAILS:

**DATASET:**
- **QEEG dataset:** 104 subjects (52 PTSD, 52 healthy)
- **EEG features:** 1140 PSD/FC values across 6 bands and 19 channels

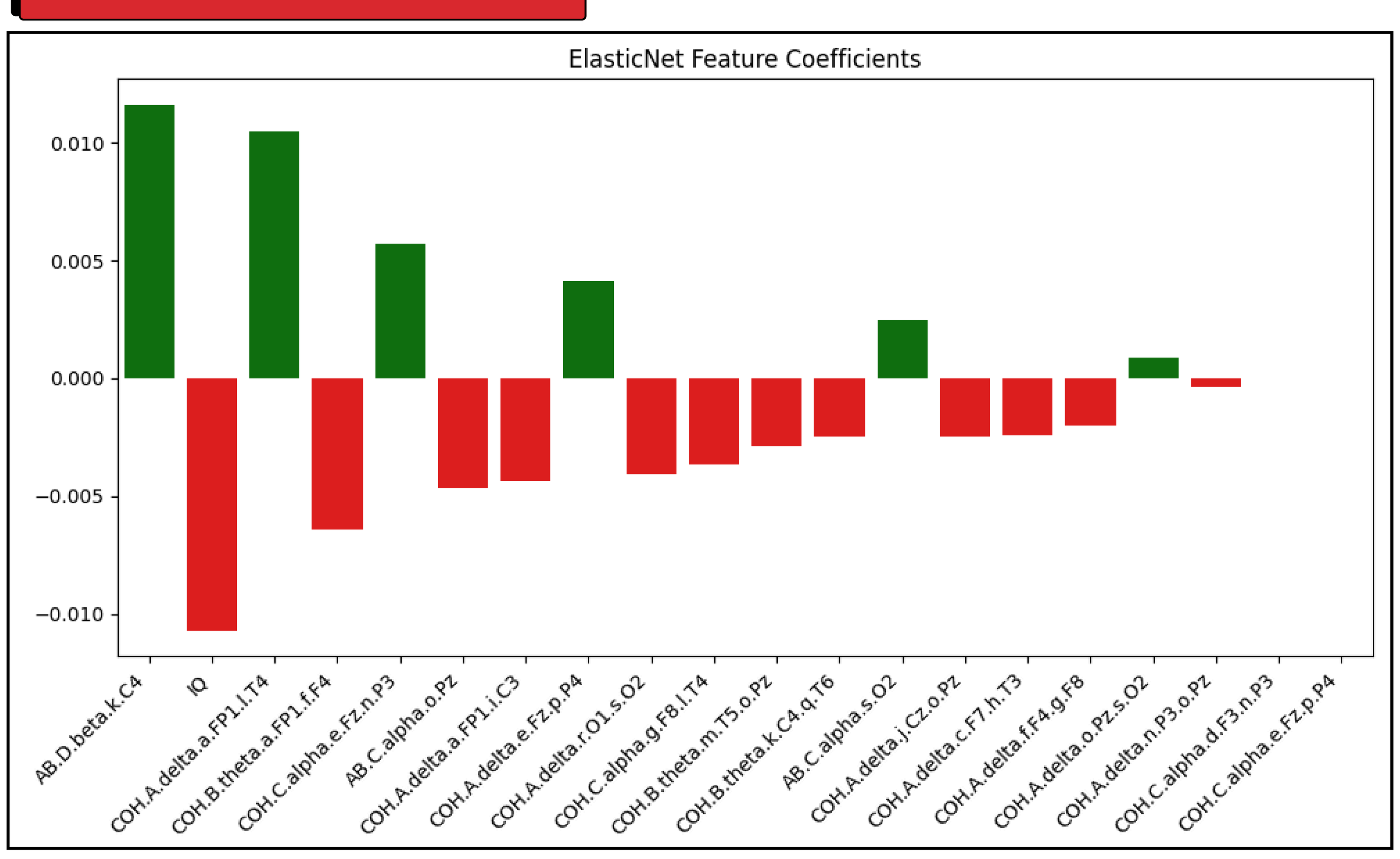
**PREPROCESSING:**
- Removed irrelevant fields
- Imputed missing values, encoded categories

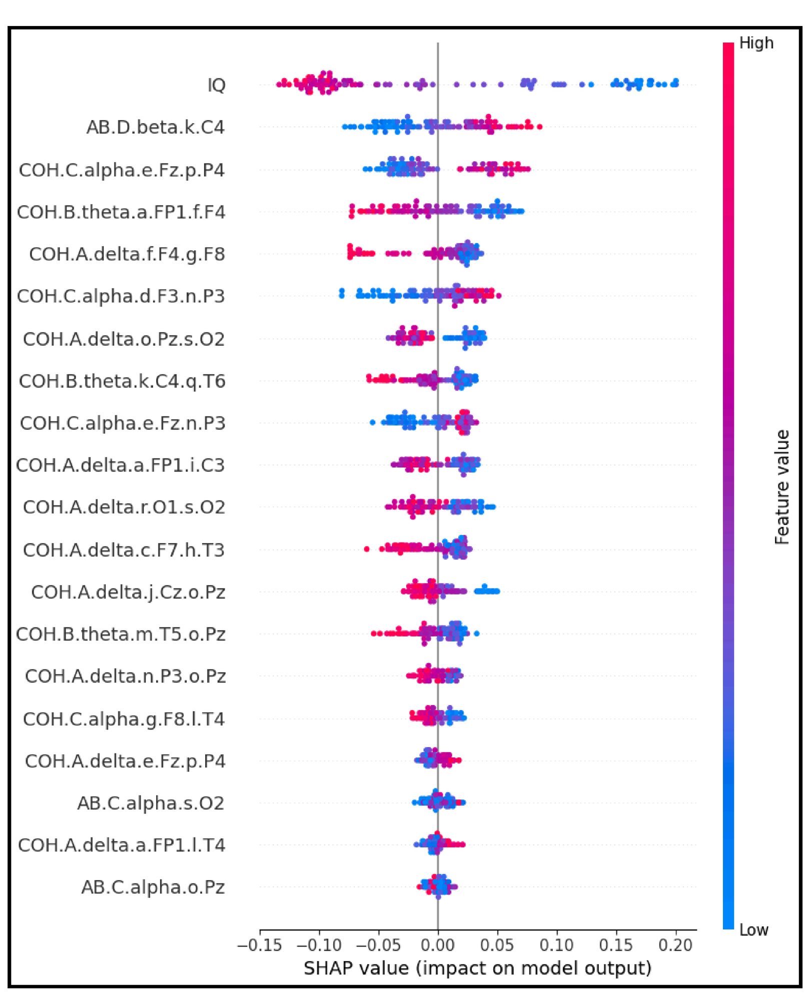
**FEATURE SELECTION:**
- Applied ElasticNet to reduce features by ~97%
- Preserved only interpretable, relevant EEG and demographic features

**MODELING:**
- Trained 7 ML models (SVC, KNN, RF, LGBM, XGBoost, AdaBoost, CatBoost)
- Tuned using Optuna Optimisation

**EXPLAINABLE AI (XAI):**
- Used SHAP for global insights
- LIME for local interpretations
- ELI5 for feature importance ranking

## TEST SET-UP AND RESULTS:



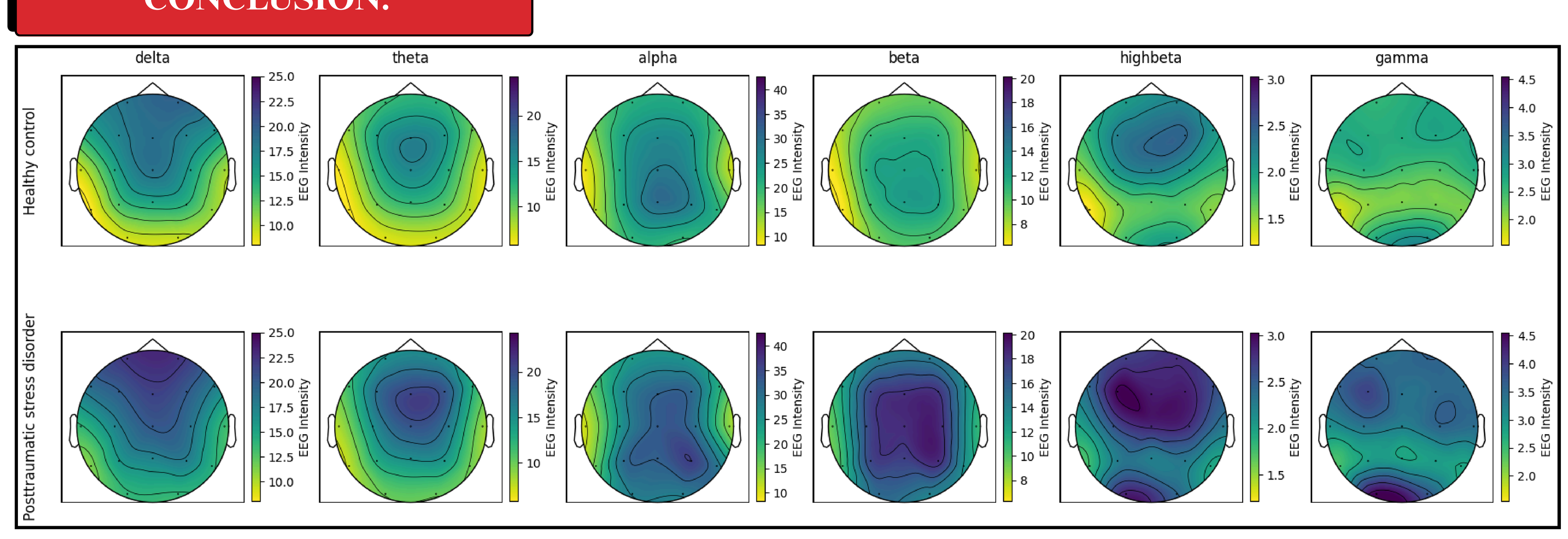



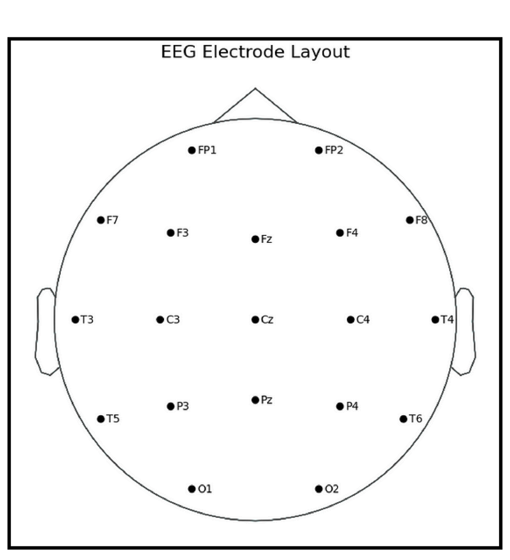

**Overview of prior work vs. proposed model**

| Study / Model | Interpretability | Interpretability |
|---|---|---|
| **Proposed Model** | **AUC: 0.85%** | Strong global/local: SHAP, LIME, ELI5 |
| **Kim et al. [2]** | Accuracy: 75.24% | Not discussed |
| **Harricharan et al. [3]** | AUC: 80.4% | Highlighted connectivity patterns in insula |
| **Salminen et al. [4]** | PTSD: 69%, ELS: 64% | Identified structural markers |
| **García-Valdez et al. [5]** | Accuracy: 0.5 – 0.9 | SHAP revealed key features across modalities |

- **ElasticNet-Based Feature Selection:** Reduced the dataset by ~97%, retaining the 20 most impactful features
- **Evaluation Metrics:** Permutation test confirming statistical significance with $p < 0.05$ for AUC score
- **Best ML Model:** RandomForest with Optuna optimisation
- **Parameters:** {'n_estimators': 131, 'max_depth': 38, 'min_samples_split': 4, 'min_samples_leaf': 2}
- **Explainability Analysis:** Identified IQ, Beta PSD at C4, and Alpha FC between F3 and P3 as top global predictors.

## CONCLUSION:




Quantitative EEG has proven to be an effective biomarker for PTSD diagnosis, but its high dimensionality can pose computational and analytical challenges. This work introduces a transparent and efficient EEG-based ML approach for PTSD diagnosis by retaining only 3% of the original EEG features using EN feature selection and integrating explainable AI. SHAP, LIME, and ELI5 consistently highlighted key EEG biomarkers (e.g., Beta PSD on PMC, Alpha coherence). Visual analyses revealed elevated Beta and High-Beta activity in PTSD patients (frontal/central regions), with reduced Delta and Alpha activity, linked to arousal and cognitive impairments. These implications are consistently noticed throughout different feature importances and XAI techniques and are even backed up by previously observed neurophysiological markers [6] [7]. This indicates that ML models paired with the XAI approach on EEG data can help obtain neurobiological biomarkers of PTSD in a non-invasive and computationally effective way.

**Future Scope:**
- Multi-disorder classification (e.g., ADHD, MDD, GAD) with XAI implications and biomarker identification.
- Utilise EEG to track treatment response over time.
- Combine EEG with ECG, neuroimaging, and biometric data to enhance PTSD prediction accuracy and interpretability.

## REFERENCES

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[2] Y.W. Kim et al., "Riemannian classifier enhances the accuracy of machine-learning-based diagnosis of PTSD using resting EEG," doi: 10.1016/j.pnpbp.2020.109960.
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[7] M. Eidelman-Rothman, J. Levy, and R. Feldman, "Alpha oscillations and their impairment in affective and post-traumatic stress disorders," doi: 10.1016/j.neubiorev.2016.07.005.

**Source code and documentation accessible via GitHub.**

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**Repository:**
- github.com/ParmarDarshan29/NeuroML-PTSD